

KARTHIKEYAN V

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A highly Analytical Mechanical Engineering graduate with a strong foundation in automotive design (BIW, Plastic, Wiring Harness) and manufacturing processes. Seeking a role to utilize skills in CATIA V5 and problem-solving to support production and design objectives and contribute to a professional team.

PROJECTS

Design and Fabrication of Safety Riding Jacket Pads Using FRP Material:

- Motorcycle riders face significant risks in accidents, making protective gear essential for safety. This project focuses on the design and fabrication of riding jacket paddings using Fiber Reinforced Plastic (FRP) materials to enhance protection and durability.
- The use of FRP improves the impact resistance and performance of the pads while maintaining flexibility and comfort. This innovation has the potential to improve rider safety while providing a cost-effective alternative for protective gear.

Design and Fabrication of Pneumatically Enhanced Prosthetic

- Developed and presented a research project on pneumatic prosthetics, highlighting their potential to improve mobility, dexterity, and comfort for amputees through compressed air systems. Addressed key advantages over traditional prosthetics, including lightweight design and enhanced control.
- Identified challenges such as high manufacturing costs and complex control systems. Proposed future advancements in materials, control technology, and manufacturing to improve accessibility and affordability.

PROFESSIONAL EXPERIENCE

QUALITY INSPECTOR | TVS Sundram fasteners | Jan 2026 – Present

- In my role, I inspect different automotive parts such as gears, pinions, wheel hub and other machined components. I check the parts during production and after completion to make sure they match the drawing and required size and tolerances. I also maintain inspection records and follow quality and safety procedures for quality parts and less rejection rate of components.
- I work on inspection equipment such as CMM (Coordinate Measuring Machine), Contour Machine, Profile Projector, Flatness Projector, and Roughness Tester. I use them to check dimensions, profile accuracy, flatness, and surface roughness of gears, pinions, and other precision parts to ensure they meet drawing specifications.

WORKSHOP

- Attended workshop on “Build and Fly Your Own Drone” in Puducherry Technological University by AIC-PEC FOUNDATION.
- Attended workshop on “Drone Technology” in Ignite Skylabs (IGSL), Manakula Vinayagar Institute of Technology.

SKILLS

- **Software Tools:** Catia V5, Siemens NX, SolidWorks, AutoCAD, Java, Python
- **Communication:** English, Tamil

CERTIFICATIONS

Automotive Sheet Metal Design | Skill-Lync

- Proficient in designing automotive Body in White (BIW) components using CATIA V5. Gained practical experience creating parts like hoods and fenders, with an understanding of industry-specific standards and manufacturing constraints.

Automotive Plastic Design | Skill-Lync

- Skilled in automotive plastic component design utilizing CATIA V5, with a focus on injection modelling principles. Applied knowledge to design parts such as car door panel, incorporating considerations for manufacturability and material properties.

Geometric Dimensioning and Tolerancing (GD&T) | Skill-Lync

- Acquired expertise in Geometric Dimensioning and Tolerancing (GD&T) to precisely define and control feature variations. Applied GD&T symbols and principles to ensure functional fit and ease of manufacturing for complex components.

Automotive Wiring Harness using CATIA V5 | Skill-Lync

- Specialized in Automotive Wiring Harness Design using CATIA V5 Electrical Harness. Skilled in 3D geometrical bundle routing, component placement (connectors, clips), and performing clearance analysis. Proficient in generating detailed 2D manufacturing drawings via Harness Flattening.

Automotive BIW Fixture Design using CATIA V5 | Skill-Lync

- I successfully **developed and modelled essential welding and assembly fixtures** for the vehicle's primary structure (**Body in White**) using **CATIA V5**. I designed key holding components (Pin, Clamp, and Rest units) based on industrial **principles** to ensure precision.

EDUCATION

Bachelors of Engineering – Mechanical Engineering

Sri Manakula Vinayagar Engineering College

CGPA:7.84

2021 – 2025

Grade XII

Aditya Vidyashram (CBSE), Puducherry

75%

2021

Grade X

Aditya Vidyashram (CBSE), Puducherry

63%

2019

ADDITIONAL INFORMATION

Personal Traits: Collaborative team player, Learning agility, Results oriented problem solver

Hobbies: Learning personal finance strategies, playing chess, Following tech news